



ELEMENTARY MATH (CALCULUS)



DIFFERENTIATION: FORMULAE, RULES & EXAMPLES

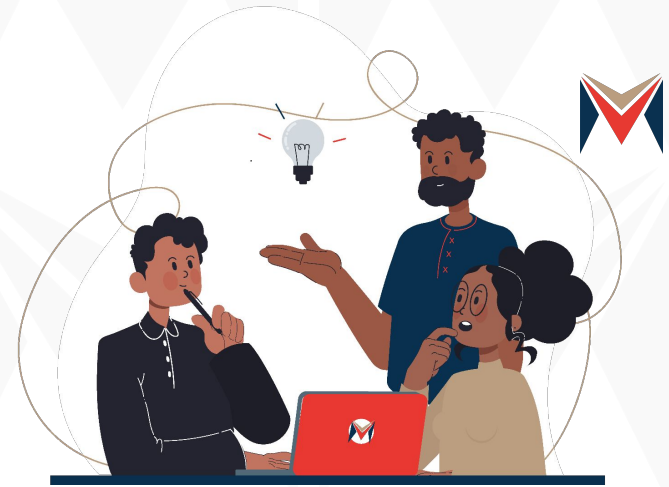
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MSc Mathematics



LEARNING OBJECTIVES

Upon completion of this unit, you should be able to:

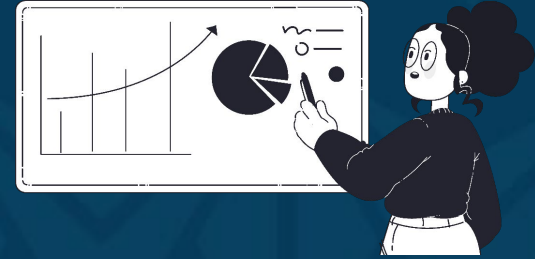
- identify the rules of differentiation,
- know the different formulas for finding derivatives,
- solve different examples on differentiation.





INTRODUCTION

In calculus, the derivative is a fundamental mathematical concept that measures how a function changes as its input (independent variable) changes. Geometrically, it represents the slope or rate of change of a function at a specific point. The derivative is a central tool in calculus and plays a crucial role in understanding motion, rates of growth, optimization, and more.





POWER RULE

Differentiate $f(x) = x^4$

Solution:

The power rule states that when $f(x) = x^n$

$$f'(x) = nx^{n-1}$$

From the given question,

$$f'(x) = 4x^{4-1}$$

Hence,

$$f'(x) = 4x^3$$





POWER RULE

Find the derivative of $f(x) = 3x^{100} - 4x$

Solution:

The power rule states that when $f(x) = x^n$

$$f'(x) = nx^{n-1}$$

From the given question,

$$\begin{aligned} f'(x) &= (3 \times 100)x^{100-1} - (4 \times 1)x^{1-1} \\ &= 300x^{99} - 4x^0 \\ &= 300x^{99} - 4 \end{aligned}$$

where, $x^0 = 1$





PRODUCT RULE

Find $f'(x)$ of $f(x) = x^2 \cdot \sin x$

Solution:

The product rule states that when $f(x) = u(x) \cdot v(x)$

we have,

$$f'(x) = u(x)v'(x) + u'(x)v(x)$$

From the given question,

$$u(x) = x^2, v(x) = \sin x, u'(x) = 2x, v'(x) = \cos x$$

$$f'(x) = x^2 \cos x + 2x \sin x$$





PRODUCT RULE

Find the first derivative of $f(x) = (4x + 6)(2x^3 - 5x)$

Solution:

The product rule states that when $f(x) = u(x) \cdot v(x)$

$$f'(x) = u(x)v'(x) + u'(x)v(x)$$

From the given question,

$$\begin{aligned}u(x) &= (4x + 6), v(x) = (2x^3 - 5x) \\ , u'(x) &= 4, v'(x) = 6x^2 - 5 \\ f'(x) &= (4x + 6)(6x - 5) + (2x^3 - 5x)(4)\end{aligned}$$

or simplify to get,

$$f'(x) = 8x^3 + 24x^2 - 32x - 30$$





QUOTIENT RULE

Differentiate $f(x) = \frac{x}{\cos(x)}$

Solution:

The quotient rule states that given $f(x) = \frac{u(x)}{v(x)}$

Then, $f'(x) = \frac{v(x)u'(x) - u(x)v'(x)}{(v(x))^2}$

From the given question,

$$u(x) = x, v(x) = \cos(x), u'(x) = 1, v'(x) = -\sin(x)$$

$$f'(x) = \frac{\cos(x) \cdot 1 - x \cdot (-\sin(x))}{(\cos(x))^2}$$

$$f'(x) = \frac{\cos(x) + \sin(x)}{\cos^2(x)}$$





QUOTIENT RULE

Find the derivative of $f(x) = \frac{(6x-3)}{5x^2}$

Solution:

The quotient rule states that given $f(x) = \frac{u(x)}{v(x)}$

Then, $f'(x) = \frac{v(x)u'(x) - u(x)v'(x)}{(v(x))^2}$

From the given question,

$$u(x) = (6x - 3), v(x) = 5x^2, u'(x) = 6, v'(x) = 10x$$

$$f'(x) = \frac{5x^2 \cdot 6 - (6x - 3) \cdot 10x}{(5x^2)^2}$$

$$f'(x) = \frac{30x^2 - 60x^2 + 30x}{25x^4} = \frac{-30x^2 + 30x}{25x^4}$$

$$f'(x) = \frac{6(1 - x)}{5x^3}$$





CHAIN RULE

What is the first derivative of $f(x) = \cos(x^2)$

Solution:

The chain rule states that when $f(x) = f(g(x))$

if $y = f(u)$, where $u = g(x)$

Then, $f'(x) = y'(u) \cdot u'(x)$

From the given question,

if $y = \cos(x^2)$, then $y = \cos(u)$
 $y'(u) = -\sin(u)$, and $u'(x) = 2x$,

$$f'(x) = -\sin(u) \cdot 2x = -\sin(x^2) \cdot 2x$$

Therefore, $f'(x) = -2x\sin(x^2)$





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TASK

What is the first derivative of $f(x) = (2x - 4)^2$





IMPLICIT FUNCTION

Differentiate $x^2 + y^2 = 25$

Solution:

We have to differentiate both sides with respect to x

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(25)$$

Applying the chain rule for y^2 :

$$2x + 2y \frac{dy}{dx} = 0$$

Isolate $\frac{dy}{dx}$:

$$\begin{aligned}\frac{dy}{dx} &= \frac{-2x}{2y} \\ \frac{dy}{dx} &= \frac{-x}{y}\end{aligned}$$





IMPLICIT FUNCTION

Differentiate $xy + 6x^4 = y$

Solution:

We have to differentiate both sides with respect to x

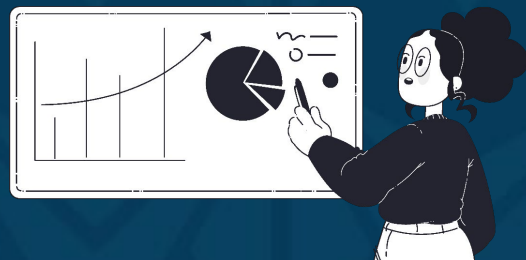
$$\frac{d}{dx}(xy + 6x^4) = \frac{d}{dx}(y)$$

$$1. y + x \frac{dy}{dx} + 24x^3 = \frac{dy}{dx}$$

$$x \frac{dy}{dx} - \frac{dy}{dx} = -24x^3 - y$$

Isolate $\frac{dy}{dx}$:

$$\begin{aligned} \frac{dy}{dx}(x - 1) &= -(24x^3 + y) \\ \therefore \frac{dy}{dx} &= \frac{-(24x^3 + y)}{(x - 1)} \end{aligned}$$





PARAMETRIC DIFFERENTIATION

Consider the parametric equations: $x(t) = 2t + 1$,
 $y(t) = t^2 - t$

Solution:

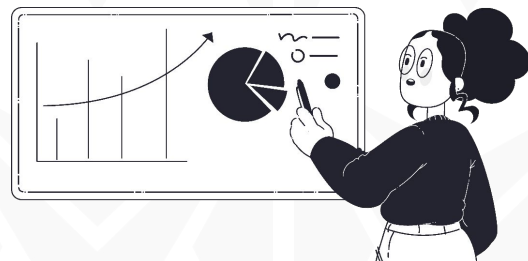
First differentiate x and y with respect to t ,

$$x'(t) = 2, y'(t) = 2t - 1$$

From the parametric rule,

$$\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$$

$$\frac{dy}{dx} = \frac{2t - 1}{2}$$

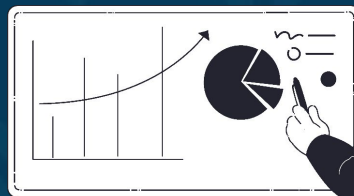




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TASK

Consider the parametric equations: $x(t) = t^4$,
 $y(t) = 7t^2 - 5t$





CONSTANT RULE

Find the derivative of $y = 5$

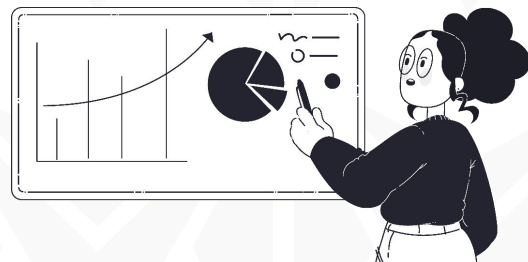
Solution:

Since the function has no variable, it is a constant.

Therefore, $\frac{dy}{dx} = 0$

NOTE:

The derivative of a constant is zero.





THE FIRST PRINCIPLE

Differentiate $y = 3x$ using the first principle.

Solution:

Recall the formula, $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

where, $f(x) = 3x$

$$f'(x) = \lim_{h \rightarrow 0} \frac{3(x+h) - 3x}{h}$$

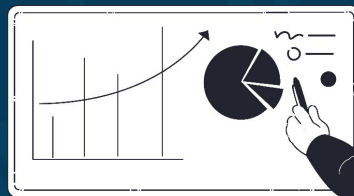
$$f'(x) = \lim_{h \rightarrow 0} \frac{3x + 3h - 3x}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{3h}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} 3$$

Therefore,

$$y' = 3$$





THE FIRST PRINCIPLE

Differentiate $y = \sqrt{x}$ using the first principle.

Solution:

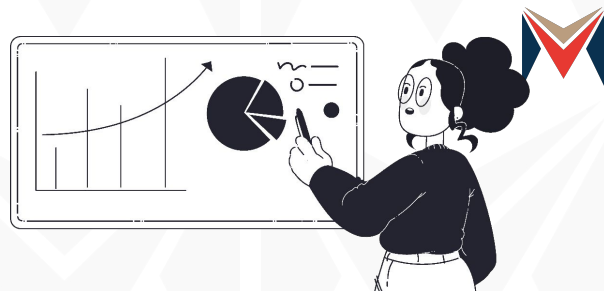
Recall the formula, $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

where, $f(x) = \sqrt{x}$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(\sqrt{x+h} - \sqrt{x})}{h} \cdot \frac{(\sqrt{x+h} + \sqrt{x})}{(\sqrt{x+h} + \sqrt{x})}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h) - (x)}{h(\sqrt{x+h} + \sqrt{x})}$$





THE FIRST PRINCIPLE (CONT'D)

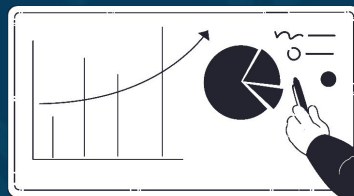
$$f'(x) = \lim_{h \rightarrow 0} \frac{x + h - x}{h(\sqrt{x+h} + \sqrt{x})}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{x+h} + \sqrt{x})}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{1}{(\sqrt{x+h} + \sqrt{x})}$$

$$f'(x) = \frac{1}{\sqrt{x+0} + \sqrt{x}}$$

$$f'(x) = \frac{1}{2\sqrt{x}}$$





CONSTANT MULTIPLE RULE

Differentiate $y = \sqrt[5]{x^3}$ with respect to x .

Solution:

$\sqrt[3]{x^5}$ can also be written as $x^{\frac{3}{5}}$

Using the power rule,

$$y' = nx^{n-1}$$

$$y' = \frac{3}{5}x^{\frac{3}{5}-1}$$

$$y' = \frac{3}{5}x^{-\frac{2}{5}}$$

$$\text{or } y' = \frac{3}{5\sqrt[5]{x^2}}$$





PARTIAL DIFFERENTIATION

Consider $f(x, y) = 3x^2 + 2xy - y^3$. Find the

partial derivatives, $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

Solution:

For $\frac{\partial f}{\partial x}$ we treat y as a constant.

$$\frac{\partial f}{\partial x} = 6x + 2y$$

For $\frac{\partial f}{\partial y}$ we treat x as a constant.

$$\frac{\partial f}{\partial y} = 2x - 3y^2$$



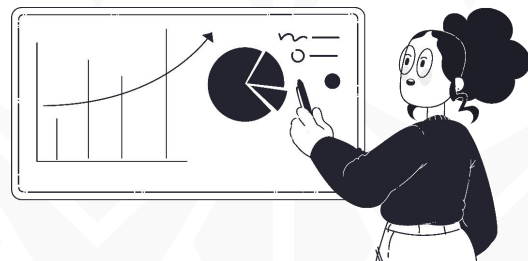


SUMMARY

The derivative of $f(x)$ with respect to x is denoted as $f'(x)$, y' , $\frac{dy}{dx}$, $\frac{df}{dx}$, $\dot{y}$ (where y is a function of x).

The power rule can be considered as the general rule of differentiation while the first principle is called the "limit definition of the derivative".

There are various rules and formulas for finding derivatives of products, quotients, and compositions of functions.





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FURTHER READING

"Calculus" by James Stewart.

"Calculus: Early Transcendentals" by Howard Anton, Irl Bivens, and Stephen Davis.

"Calculus" by Ron Larson and Bruce H. Edwards.





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THANK YOU